

FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

1. Case Details

Case ID	CASE-1630D946 1630d946-2c93-4c50-8a66-3b7f07bb0e2b
Submitted by	testreviewer L!
Submission date	2026-10-07 06:44 UTC
Documents	1 Types: Vendor invoice
Case type	School document
Status / decision	Submitted / Pending review
Report generated	2026-10-07 06:45 UTC by testreviewer L! (Reviewer L1)
Report ID	d01bf51d-1a46-4518-b89d-ff3fe6309a9c

Preliminary report. Automated analysis had not finished when this report was generated. Still outstanding: Field validation of 'caseimage1.pdf'; Issuer verification of 'caseimage1.pdf'; Metadata forensics of 'caseimage1.pdf'; Error level analysis of 'caseimage1.pdf'; Copy move detection of 'caseimage1.pdf'; Duplicate detection of 'caseimage1.pdf'.

2. Executive Summary

Not yet scored. The risk engine has not produced an assessment for this case, so there is no overall tier.

Why: No risk tier yet — automated analysis is still in progress.

Decision status: Pending review.

Exceptions: 2 item(s) did not pass, 2 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

3. Explainable Findings

The case has not been scored, so there are no triggered risk reasons yet.

4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — caseimage1.pdf (Vendor invoice · 1 passed, 1 flagged)

Check	Result	Details
Font consistency Text set in a different font family from the text around it (text layer, or OCR font recognition on scans).	Pass	No anomaly detected by this check.
Visual inconsistency / AI-generation review Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation.	Flag	2 finding(s). Page 1 — Font consistency: The Arabic text at the top center is in a different font style and color (pink) compared to the rest of the document which uses a standard sans-serif font in black or blue. Also, the 'Remarks / Description:' section us...

5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (2 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

No.	Exception	Where	Highlight
1	Visual review: font consistency inconsistency (model judgment) Page 1 — Font consistency: The Arabic text at the top center is in a different font style and color (pink) compared to the rest of the document which uses a standard sans-serif font in black or blue. Also, the 'Remarks / Description:' section uses a different font color (yellow background with red text) compared to th...	Document 1 (caseimage1.pdf) · page 1	R1
2	Visual review: color / contrast consistency inconsistency (model judgment) Page 1 — Color / contrast consistency: The 'Remarks / Description:' box has a yellow background that contrasts with the plain white background of the rest of the page. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding.	Document 1 (caseimage1.pdf) · page 1	R2

6. Limitations and Assumptions

Please read this section before relying on anything in this report.

Limitations of This Analysis

What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

Three kinds of finding

Deterministic pixel and structure checks — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

Deterministic rule-based field checks — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

Model-judgment checks — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

Absence of a finding

The absence of a flagged finding does not guarantee that a document is genuine. It means only that no supported check detected an anomaly.

Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

Check	Status	Coverage on this case
Metadata forensics	Did not run	Did not run on any of the 1 document(s).
Font consistency	Ran	Ran on 1 of 1 document(s); 0 flagged.
Error level analysis (ELA)	Did not run	Did not run on any of the 1 document(s).
Copy-move detection	Did not run	Did not run on any of the 1 document(s).
Visual inconsistency / AI-generation review	Ran	Ran on 1 of 1 document(s); 1 flagged.
Signature / stamp detection	Did not run	Did not run on any of the 1 document(s).
Duplicate detection	Did not run	Did not run on any of the 1 document(s).
Field validation	Did not run	Did not run on any of the 1 document(s).
Issuer verification	Did not run	Did not run on any of the 1 document(s).
Signature comparison (advisory)	Did not run	Did not run — no reference signature was set for this case.
Cross-document consistency	Not applicable	Not applicable — this case has a single document.

Risk score and rule versions

This case had not been scored when the report was generated, so no risk-rule versions apply.

Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: caseimage1.pdf: Due date (70% confidence).
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.



7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

Time	Event	Actor	Detail
2026-10-07 06:44 UTC	Case created	testreviewer L! (Reviewer L1)	
2026-10-07 06:44 UTC	Document uploaded	testreviewer L! (Reviewer L1)	caseimage1.pdf; SHA-256 1d2ff2211bce7ff...
2026-10-07 06:45 UTC	Visual inconsistency review completed	System	caseimage1.pdf
2026-10-07 06:45 UTC	Font consistency check completed	System	caseimage1.pdf
2026-10-07 06:45 UTC	Text extraction, classification and field extraction completed	System	caseimage1.pdf

8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

Document 1 — caseimage1.pdf

Field	Value	Confidence	Located
Issuer	PINK & BLUE NURSERY METTE حضانة الوردي والأزرق	90%	p.1
Reference number	MLG15112257877	95%	p.1
Date	2027-11-15	95%	p.1
Amount	10,300.00 AED	90%	p.1
Subtotal	10,300.00 AED	90%	p.1
Tax rate	Not found	uncertain	
Tax amount	0.00 AED	90%	p.1
Bill to student details	Hamdan Khān Proand Buy A Nhưbạc B1	90%	p.1
Due date	15 No/ 2027	70% uncertain	p.1
Vat ch	0.00	90%	p.1

8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

Document	SHA-256	Verification
caseimage1.pdf 12,479 bytes	1d2ff2211bce7f7ff163213652d0996c612a2aee9e59c632a6419b176f3c86bb	VERIFIED

8.3 Technical parameters used by each check

Page rendering (input to ELA, copy-move, duplicate, visual review)

Render resolution	200 DPI
Colour model	RGB, converted to BGR for OpenCV

Visual inconsistency / AI-generation review

Model	gpt-4.1-mini
Independent runs per page	2
A finding is kept if	both runs report it, or one reports it with high confidence
Image sent to model	page render, long edge ≤ 1536px

Field locations (purple highlights)

Method	extracted values matched back onto OCR word positions (Azure Document Intelligence Layout)
Repeated values	resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)
Accuracy	approximate; a field that cannot be matched is reported without a highlight

9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

Colour key

	ELA (solid red)
	Copy-move (solid orange)
	Font mismatch in the text layer (solid fuchsia)
	Field exception (solid purple)
	Model judgment (dashed blue, approximate)
	Page-level finding (not localized)

Region index

Region	Document	Page	What it shows
R1	Document 1 (caseimage1.pdf)	1	Visual review — font consistency (approximate)
R2	Document 1 (caseimage1.pdf)	1	Visual review — color / contrast consistency (approximate)

Section 9 · Document 1 of 1 · regions R1, R2

 Model judgment (dashed blue, approximate)

PINK & BLUE NURSERY

AL ALAM - UAE

حضانة الوردى والأزرق

R1 · Visual review — font consistency (approximate)

TAX INVOICE

Invoice #: MLC15112257877
Date: 15-Nov-2022
Due Date: 15-Nov-2022

BILL TO / STUDENT DETAILS

Hamdan Khan
Pink and Blue Al Khabeesi - Br1

DESCRIPTION	QTY	UNIT PRICE (AED)	SUBTOTAL (AED)
Tuition fee	1	10,300.00	10,300.00
Registration	0	0.00	0.00
Transportation	0	0.00	0.00
Uniform	0	0.00	0.00

Remarks / Description:
Monthly Payment Fee (Partial)
*** Note: Registration is Non-Refundable ***

Subtotal	10,300.00
VAT (5%)	0.00

Total Amount Due10,300.00 AED

R2 · Visual review — color / contrast consistency (approximate)

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Page 1 of 2

Page 1 — Font consistency: The Arabic text at the top center is in a different font style and color (pink) compared to the rest of the document which uses a standard sans-serif font in black or blue. Also, the 'Remarks / Description:' section uses a different font color (yellow background with red text) compared to the main body text. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal...

Page 1 — Color / contrast consistency: The 'Remarks / Description:' box has a yellow background that contrasts with the plain white background of the rest of the page. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding.